# Portfolio Investment | Strategy Backtesting

FN6901 Group Assignment • U.S. Sector ETFs • November 1999–October 2021

## Five portfolios. One historical sample. Risk, return and portfolio construction.

Source: Portfolio Investment Assignment, rendered 13 October 2026

# Research question and design

What changes when we rebalance or select sectors using factor-model alpha?

- Nine SPDR sector ETFs; SPY is the market benchmark
- Adjusted-close daily returns, 1999-11-01 to 2021-10-08
- Equal-weight buy-and-hold, quarterly and annual rebalancing
- Fama–French three-factor alpha: monthly top-five selection
- No transaction costs; historical in-sample backtest

# Backtest mechanics

Comparable assumptions, different portfolio decisions

- Buy & Hold: invest 1/9 in each sector ETF at inception
- Quarterly: reset to 1/9 every 63 trading days
- Annual: reset to 1/9 every 252 trading days
- FF Alpha Top 5: estimate alpha from 63 trailing daily observations
- Every 21 trading days select five ETFs at 20% each; weights apply next day

# Performance at a glance

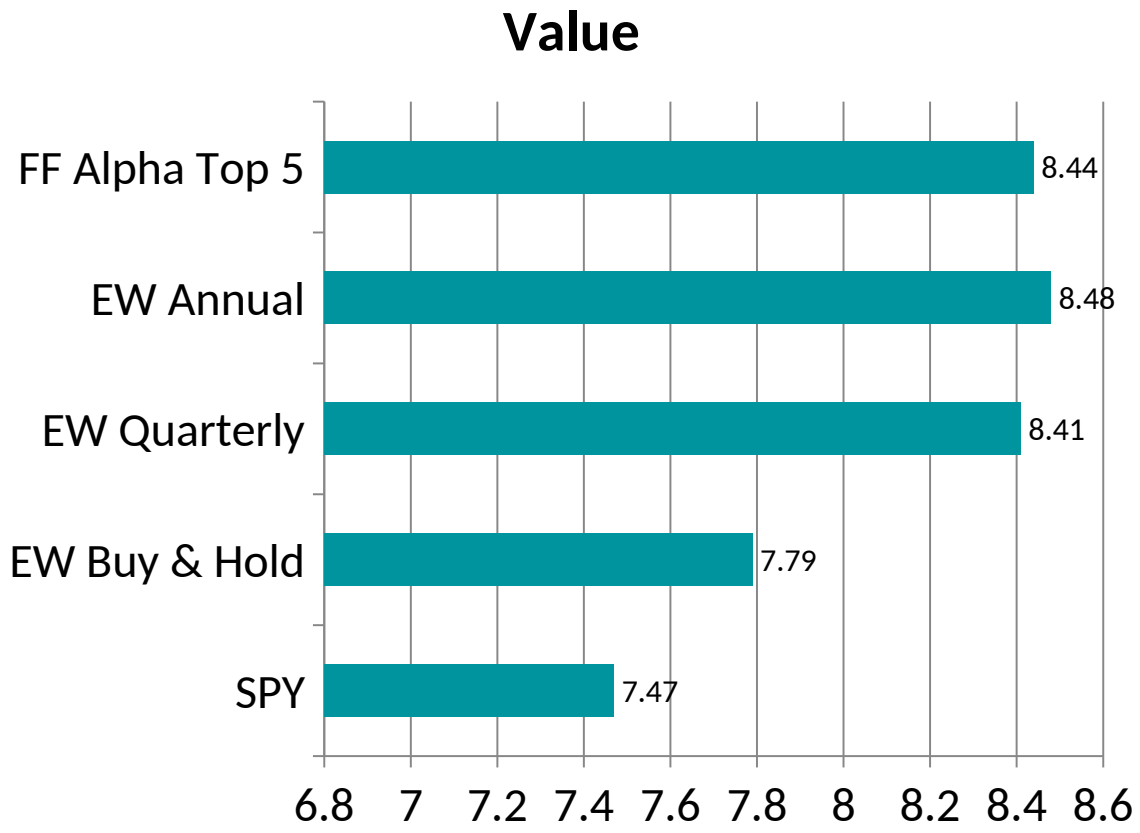
Full sample; Sharpe = CAGR / annualized daily volatility (risk-free rate assumed zero)

| Strategy | CAGR | Volatility | Sharpe | Max DD | Final $1 |
|---|---|---|---|---|---|
| SPY | 7.47% | 19.64% | 0.3805 | −55.20% | 4.8480 |
| EW Buy & Hold | 7.79% | 18.68% | 0.4173 | −52.32% | 5.1743 |
| EW Quarterly | 8.41% | 18.70% | 0.4497 | −52.98% | 5.8616 |
| **EW Annual** | **8.48%** | **18.59%** | **0.4563** | **−52.86%** | **5.9498** |
| FF Alpha Top 5 | 8.44% | 19.02% | 0.4437 | −50.95% | 5.8956 |

**Annual rebalancing leads in CAGR and Sharpe; FF Top 5 has the smallest maximum drawdown.**

# Return and risk-adjusted performance

Annual rebalancing leads by a narrow margin




**8.48%  Annual EW CAGR**

**0.4563  Annual EW Sharpe**

**8.44%  FF Top 5 CAGR**

**0.4437  FF Top 5 Sharpe**

Bars show annualized return in percent.

# Equity curves | Growth of $1

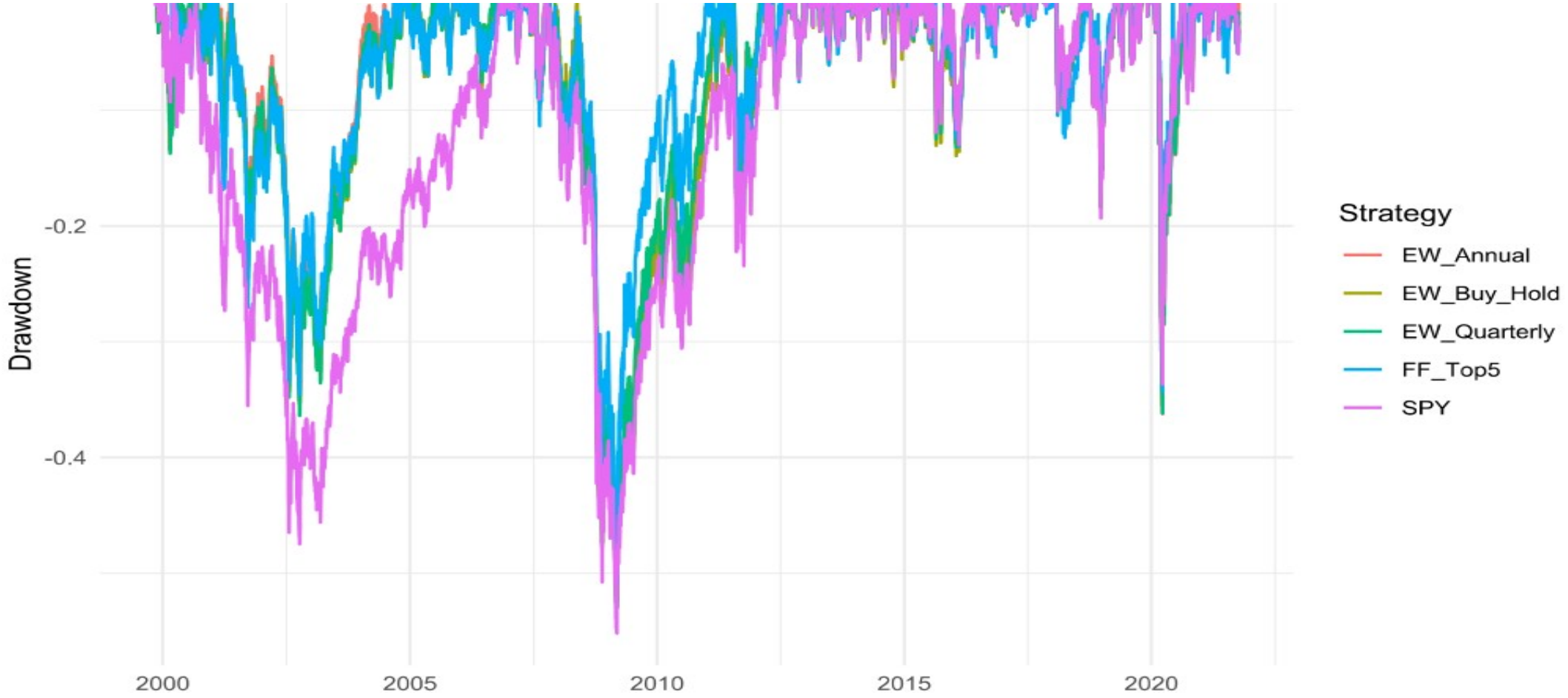
Rebalanced and alpha-selected portfolios finish above buy-and-hold and SPY




Source: rendered assignment, page 17.

# Drawdown | Downside experience

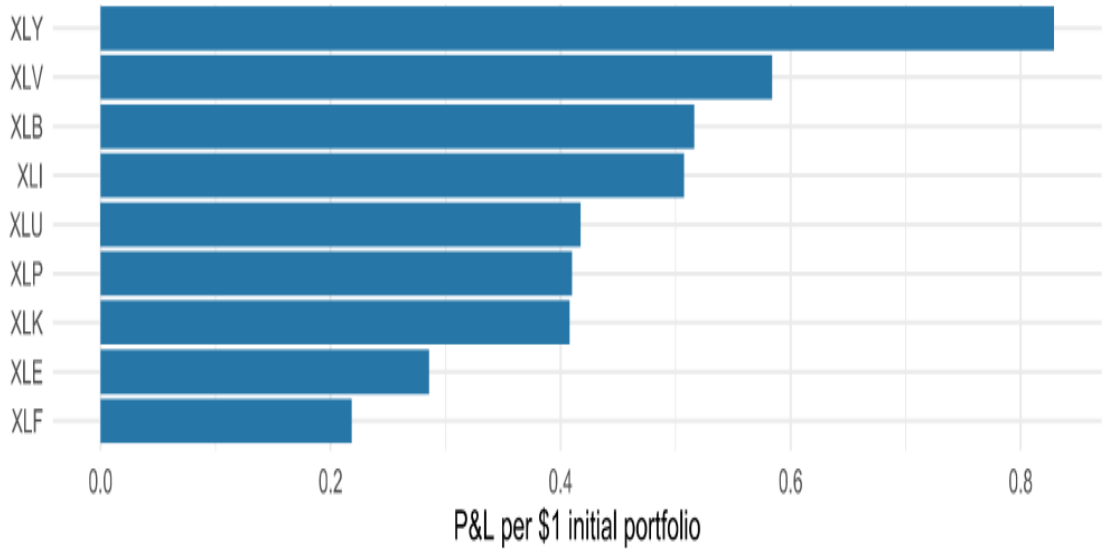
Every strategy suffered deep losses; FF Top 5 has the least severe maximum drawdown

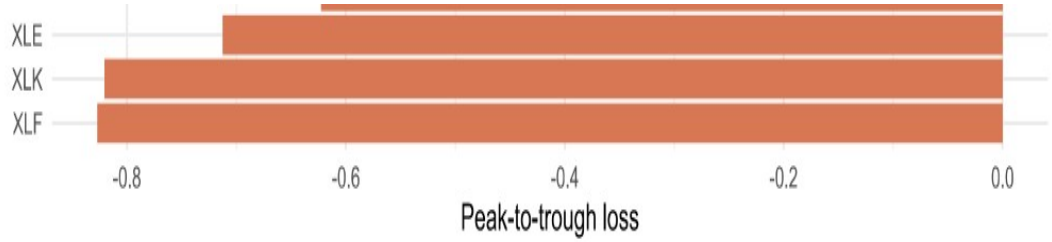


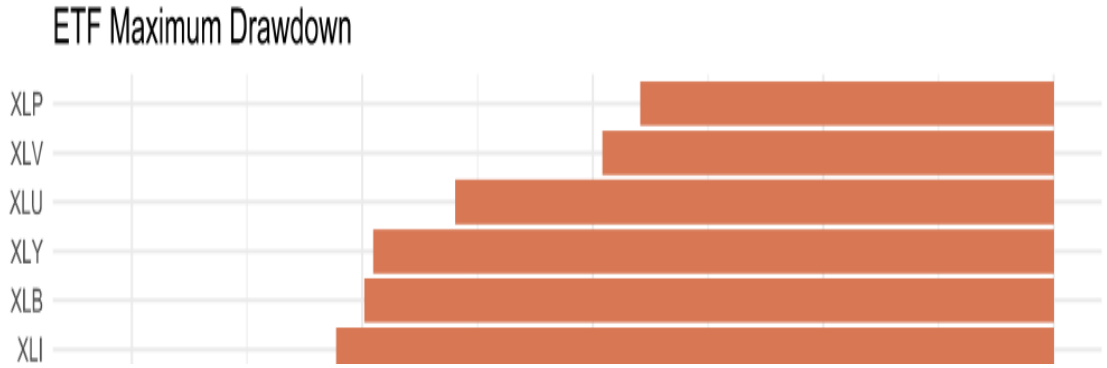

Maximum drawdowns range from −50.95% to −55.20%. Source: report page 18.

# Sector contributions and standalone risk

Buy-and-hold P&L leaders are not necessarily the lowest-risk sectors




## 3. EW with Re-balance

Re-balance is an action to reset the weights so the allocation into each ETF has equal market value.

Without re-balance, some ETFs can grow their value to a certain extent that reduces the diversification of our investment portfolio.

- The starting allocation for each ETF is 1/9.
- On the next re-balance day, calculate the value of the portfolio. For example, it has grown to 1.3 based

XLY and XLV lead P&L contribution; XLF and XLK show the worst individual drawdowns.

# Fama–French alpha selection

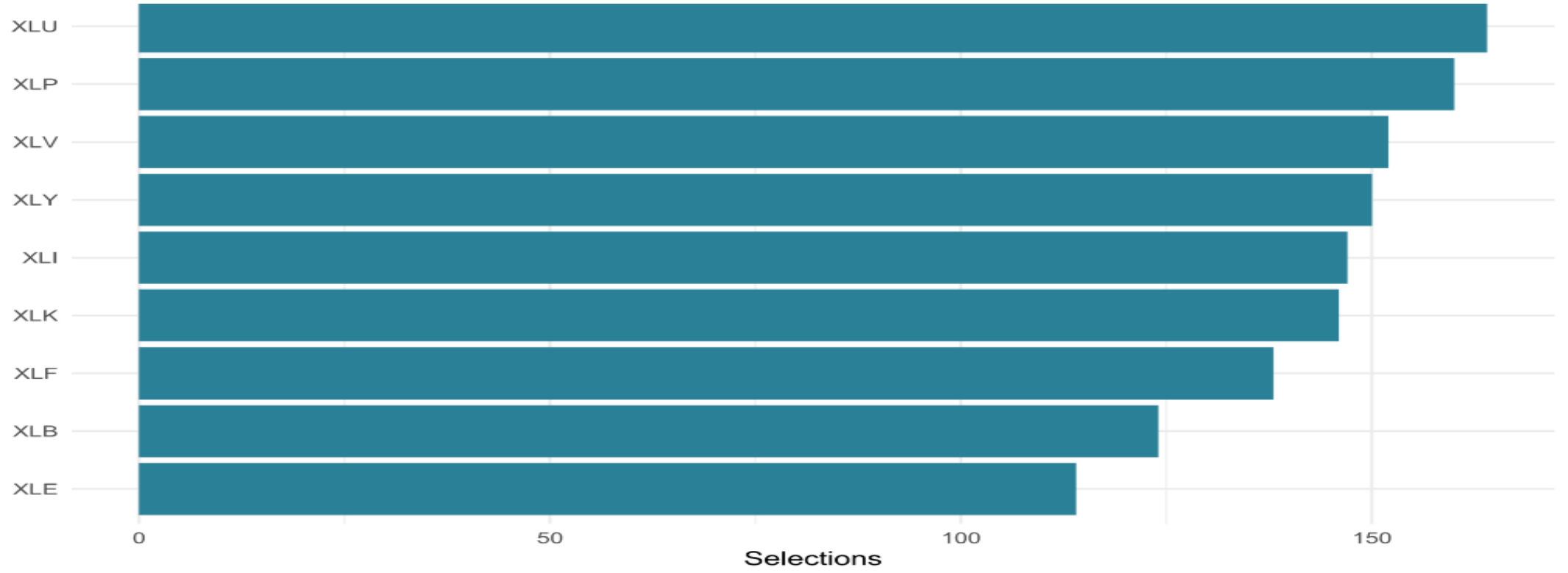
A rules-based, backward-looking signal—not a guaranteed forecast

$$\text{ETF excess return} = \alpha + \beta M \times \text{Mkt-RF} + \beta S \times \text{SMB} + \beta H \times \text{HML} + \varepsilon$$

- Estimate alpha using the previous 63 trading days
- Rank the nine ETFs by estimated alpha; choose the top five
- Rebalance every 21 trading days, with next-day implementation
- First valid rebalance on return observation 81; hold cash before investment
- Selection frequency shows persistent but changing sector preferences

# Which sectors does alpha select most often?

Historical selection frequency, not a measure of future alpha




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ggsave("ff_selection_frequency.png", p_selection, width = 9, height = 5, dpi = 180)
ggsave("etf_pnl_contribution.png", p_pnl, width = 9, height = 5, dpi = 180)
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XLU and XLP were selected most often; XLE least often in the reported sample.

# Interpretation and limitations

The key result is the comparison—not a claim of out-of-sample alpha

- Annual EW: highest CAGR (8.48%) and Sharpe (0.4563)
- FF Top 5: smallest maximum drawdown (−50.95%), but lower Sharpe
- Rebalancing may help manage drifting sector allocations
- FF alpha is estimated with a short 63-day window and may be noisy
- No fees, slippage, taxes or market impact; no out-of-sample test

# Conclusion | What would we take forward?

Portfolio rules matter; historical winners are not automatically future winners

- All four diversified strategies outperformed SPY on reported CAGR
- Annual equal-weight rebalancing offers the strongest sample Sharpe
- FF alpha selection provides a different exposure profile and drawdown outcome
- Next research step: rolling out-of-sample validation and trading-cost sensitivity

Source throughout: user-provided rendered Portfolio Investment Assignment PDF, pp. 8–20.